

Sai Srikar Chirumarri

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Education

- 2024 - 2025 **MSc. Computer Games Technology with VR**
City University of London, London, England.
- 2022 - 2024 **B.tech CSE with specialization in Robotics**
Vellore institute of technology, India.
CGPA: 8.4

Skills and Certifications

- **Programming:** C++/C, C, GLSL, Java, Python, SQL, HTML
- **Game Engines or related:** Unreal Engine, Unity, Monogame, Visual Studio
- **Tools:** Blender, Quixel mixer, Substance Painter, Maya, Marvelous Designer
- **Certifications:** Unreal Engine 5 C++ Developer, Data Structures and OOP with C++

Projects

1)The Buckerberg Files, Unity, CSharp, VR

A provocative and immersive VR experience set in a future where data privacy has become a core human right. You journey into the abandoned HQ explore, solve puzzles and uncover the truths behind the company.

I am the Gameplay programmer and level designer made in Unity where I worked in a team of 5. The tasks and features I implemented include

- i)VR player setup(VR hands, player body) and environment interactions(pockets, breaking glass, scanning keycard).
- ii)Environmental and interaction sounds, Haptics.
- iii)VR UI(Start menu, Endscreen), Implementation of different game paths

2)Shackled Shadow, Monogame, CSharp, 2D

You are a shadow who escaped the owner looking for freedom. Travel between two worlds solving puzzles, fighting enemies with your powers. Explore the world and find your freedom

I am the sole developer made with monogame. The game was built from scratch hence many game managers and systems needed to be coded they include

- i)Start and End screen with high scores, level restart etc.
- ii)An innovative game mechanic where art switches(switching world) as a powerup.
- iii)Level design extracted from txt files(Player, enemy or tile positions can be changed).
- iv)Power up drops with 2 variations, enemy movement and FSM, Input listeners, score managers.

3)Planet Clearance, Custom Game engine, C++, 3D

Third person shooter- You are a space agent who is assigned the task of cleaning the planet. Aliens have the power to teleport.

I am the sole developer made with a custom game engine. The game and the engine was built on a basic template items that I coded include

- i)Game intro screen (Quad constructed with OpenGL).
- ii)Primitive tetrahedron object made with C++ and OpenGL from scratch.
- iii)Use of audio files using a FMod file manager, A 3rd person camera controller that tracks player, and Powerups.
- iv)Special effects like fog, crossfade, explosion, UI elements, NPC FSM with 4 states.

4)Dynamic NPC System, Unreal engine, C++, VR

The project aim is to improve immersion in VR games by designing NPCs that react realistically to physical touch interactions.

I am the sole developer made in Unreal as part of my masters dissertation. The tasks and features I implemented include

- i)VR player setup(VR hands) and NPC AI with Blackboard and Behaviour tree.
- ii)A custom collision system that detects NPC body part and touch intensity.
- iii)A finite state machine that uses the new collision system with different behaviour modes(Aggressive, friendly).
- iv)Physics based animation that moves the bones using the new collision system and ragdoll physics, This dynamic system is designed to act as a plugin.